

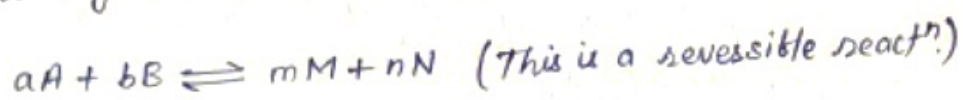
Law of Mass Action

In 1869, law of mass action was discovered by Guldberg and Waage.

STATEMENT:-

The rate of a reaction is directly proportional to the product of the active masses of reacting substances.

Active Mass :- The mass of the substance which is used during a chemical reaction. []



$$r_f \propto [A]^a \cdot [B]^b$$

Rate of forward reaction is directly proportional to the Active Mass of A and B.

$$r_b \propto [M]^m \cdot [N]^n$$

Rate of backward reaction is directly proportional to the Active Mass of M and N

$$r_f = k_f [A]^a \cdot [B]^b$$

$$r_b = k_b [M]^m \cdot [N]^n$$

where, $r_f \rightarrow$ forward reaction, $r_b \rightarrow$ backward reaction

$k_f \rightarrow$ const for forward reaction.

$k_b \rightarrow$ const for backward reaction.

At equilibrium

$$k_f [A]^a \cdot [B]^b = k_b [M]^m \cdot [N]^n$$

$$\left(\frac{\text{equilibrium const}}{\text{for reaction}} \right) K_e = \frac{k_b}{k_f} = \frac{[M]^m \cdot [N]^n}{[A]^a \cdot [B]^b}$$

It represents the law of equilibrium.

But if we write this eqⁿ in partial pressure (K_p)

$$K_p = \frac{P_M^m \cdot P_N^n}{P_A^a \cdot P_B^b}$$

But if we write this eqⁿ in molarity (K_c)

$$K_c = \frac{C_M^m \cdot C_N^n}{C_A^a \cdot C_B^b}$$